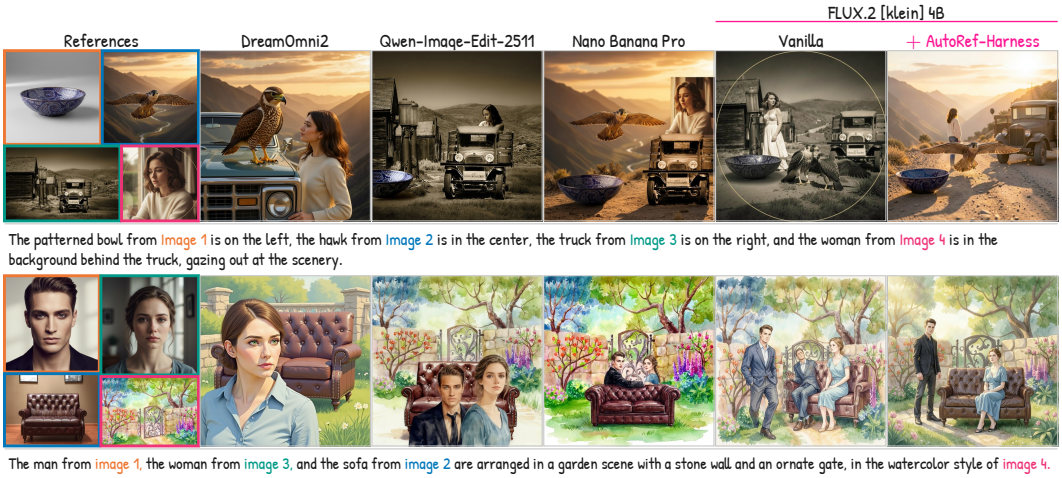



Figure 3: Qualitative comparison on held-out four-reference MultiBanana tasks. Each reference is outlined in a distinct color, and the phrase in the instruction that refers to it is shown in the same color. Baseline models often omit, duplicate, or misplace references, or copy and paste them unnaturally, whereas AutoRef-Harness preserves each reference and integrates all four into a coherent scene.

Seedream 4.5. The gains from AutoRef-Harness also transfer beyond the model used during harness optimization: applying the same harness to the larger FLUX.2 [klein] 9B improves its performance, and replacing FLUX.2 with Qwen-Image-Edit-2511 likewise yields a substantial gain. These results indicate that the benefit of the discovered harness is not limited to a particular model scale or generator family. Importantly, we achieve these improvements without updating the image generator or reasoning model; we only change the inference-time harness. Per-metric results are reported in Appendix H.1, and the harness further improves Qwen-Image-Edit-2511 after fine-tuning for multi-reference image generation with DyRef (Huang et al., 2026b; Appendix H.3).

Figure 3 illustrates qualitative examples. Baseline models often omit, duplicate, or misplace references, or paste them in unnaturally. For instance, the base FLUX.2 [klein] 4B duplicates the hawk and places the woman in the foreground rather than the background in the first example, and duplicates the man in the second. Nano Banana Pro and Qwen-Image-Edit-2511 instead produce copy-and-paste-like results in the first and second examples, respectively. AutoRef-Harness preserves each reference and naturally integrates it into the requested scene. Appendix J shows more examples.

### 6.3 Transferability of AutoRef-Harness

**Across Reference Counts.** AutoRef-Harness is discovered on four-reference MultiBanana tasks but applies to different reference counts without modification. As shown in Table 2, it consistently improves FLUX.2 [klein] 4B on both the unseen three- and five-reference settings. With three references, the resulting system outperforms all open image generators without the harness and remains competitive with proprietary models; the gain also persists in the more challenging five-reference setting. The harness likewise improves the other generators, except Qwen-Image-Edit-2511 with five references, where the generator itself fails and the harness cannot compensate (Appendix I). AutoRef-Harness thus does not rely on the four-reference structure used during harness discovery.

**Across Benchmarks and Evaluators.** We further evaluate the same AutoRef-Harness on OmniContext, which is never used during harness discovery and is scored by its official GPT-4.1 (OpenAI, 2023) evaluator rather than the Qwen3-VL-8B-Instruct evaluator used for MultiBanana. As shown in Table 3, AutoRef-Harness again improves FLUX.2 [klein] 4B and remains competitive with proprietary image models. Thus, the gains persist under simultaneous changes in both the benchmark distribution and the evaluator, without benchmark-specific modifications to the harness.

**Across Reasoning Models.** As shown in Section 6.2, AutoRef-Harness transfers across image generators of different scales and families. We next test whether the same harness also transfers across reasoning models. With the open-weight Qwen3-VL-32B (Bai et al., 2025) in place of GPT-5.5

OpenAI. Gpt-4o image generation. https://openai.com/index/introducing-4o-image-generation/, 2025a. Accessed: 2025-10-31.

OpenAI. The new chatgpt images is here. https://openai.com/index/new-chatgpt-images-is-here/, 2025b. Accessed: 2026-4-30.

OpenAI. Introducing GPT-5.5. https://openai.com/index/introducing-gpt-5-5/, 2026. Accessed: 2026-09-24.

Krista Opsahl-Ong, Michael J Ryan, Josh Purtell, David Broman, Christopher Potts, Matei Zaharia, and Omar Khattab. Optimizing instructions and demonstrations for multi-stage language model programs. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 9340–9366, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.525. URL https://aclanthology.org/2024.emnlp-main.525/.

Yuta Oshima, Masahiro Suzuki, Yutaka Matsuo, and Hiroki Furuta. Inference-time text-to-video alignment with diffusion latent beam search. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL https://openreview.net/forum?id=c9EAmyYPOv.

Yuta Oshima, Daiki Miyake, Kohsei Matsutani, Yusuke Iwasawa, Masahiro Suzuki, Yutaka Matsuo, and Hiroki Furuta. Multibanana: A challenging benchmark for multi-reference text-to-image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 448–460, June 2026.

Reid Pryzant, Dan Iter, Jerry Li, Yin Lee, Chenguang Zhu, and Michael Zeng. Automatic prompt optimization with "gradient descent" and beam search. In Houda Bouamor, Juan Pino, and Kalika Bali (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7957–7968, Singapore, December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.494. URL https://aclanthology.org/2023.emnlp-main.494/.

Xiangyan Qu, Zhenlong Yuan, Jing Tang, Rui Chen, Datao Tang, Meng Yu, Lei Sun, Yancheng Bai, Xiangxiang Chu, Gaopeng Gou, Gang Xiong, and Yujun Cai. From scale to speed: Adaptive test-time scaling for image editing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 23272–23282, June 2026.

Ruchit Rawal, Reza Shirkavand, Sayak Paul, Yuxin Wen, Heng Huang, Yizheng Chen, Tom Goldstein, and Gowthami Somepalli. Flash-bon: Instant drafts for inference-time scaling in diffusion models. In *The 19th European Conference on Computer Vision*, 2026.

Maxime Robeyns, Martin Szummer, and Laurence Aitchison. A Self-Improving Coding Agent. In *ICLR 2025 Workshop on Scaling Self-Improving Foundation Models*, 2025. URL https://openreview.net/forum?id=rShJCyLsOr.

Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22500–22510, June 2023.

Shreshth Saini, Neil Birkbeck, Yilin Wang, Balu Adsumilli, and Alan C. Bovik. Cachedsearch: Training-free cached exploration for test-time search in video diffusion, 2026. URL https://arxiv.org/abs/2607.23159.

Yongliang Shen, Kaitao Song, Xu Tan, Dongsheng Li, Weiming Lu, and Yueting Zhuang. Hugginggpt: Solving ai tasks with chatgpt and its friends in hugging face. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 38154–38180. Curran Associates, Inc., 2023. doi: 10.52202/075280-1657. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/77c33e6a367922d003ff102ffb92b658-Paper-Conference.pdf.